Nutrition and Recipe Analytics API

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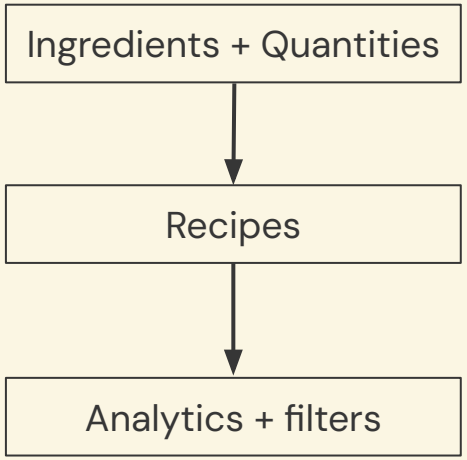
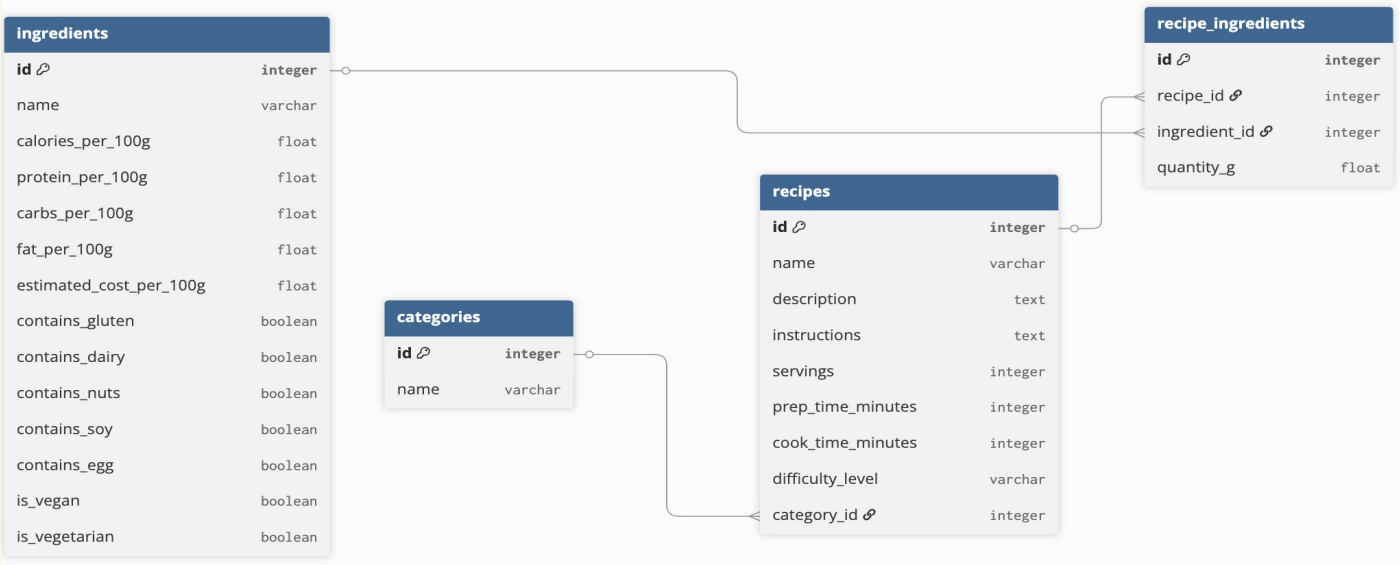
COMP3011 Web Services and Web Data

Project aim

Build a database-backed RESTful API for recipes, ingredients, and categories
Support full CRUD operations with JSON responses and correct HTTP status codes
Provide derived analytics: nutrition, allergens, cost, and difficulty
Enable practical filtering by diet, allergens, calories, cost, and preparation time



Architecture and Data Model



“recipe_ingredients resolves the many-to-many relationship and stores ingredient quantities per recipe.”

API Functionality

Core CRUD

Recipes
Ingredients
Categories
Recipe-ingredient links

Analytics

Nutrition per recipe / per serving
Allergen detection
Cost estimation
Difficulty scoring

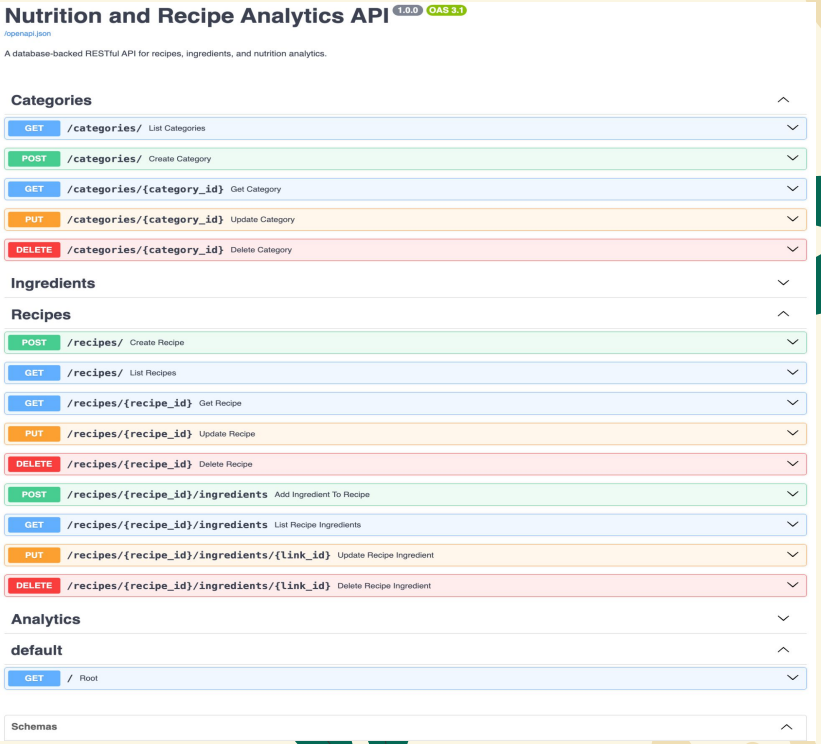
Filtering

Vegan / vegetarian
Gluten-free / dairy-free / nut-free
Max calories / max cost
Search + prep/cook time

API Design

Resource-based URLs
JSON responses
Correct HTTP methods
Clear status codes

FastAPI Swagger documentation



API Functionality

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System. This planet's name has nothing to do with the liquid metal, since Mercury was named after the Roman messenger god. Despite being closer to the Sun than Venus

Speaking of craters, many of them were named after artists or authors who made significant contributions to their respective fields. Mercury takes a little more than 58 days to complete its rotation, so try to imagine how long days must be there!

Example Outputs / Demo

vegan_only
boolean | (boolean | null)
(query)

true

Filter for vegan only recipes

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/recipes/?vegan_only=true' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/recipes/?vegan_only=true
```

Server response

Code	Details
200	

Response body


```
[
  {
    "name": "Chocolate Strawberry Bites",
    "description": "Strawberries served with dark chocolate.",
    "instructions": "Slice strawberries. Melt dark chocolate. Dip and serve.",
    "servings": 2,
    "prep_time_minutes": 10,
    "cook_time_minutes": 2,
    "difficulty_level": "easy",
    "category_id": 5,
    "id": 8
  },
  {
    "name": "Peanut Banana Toast",
    "description": "Toast with peanut butter and banana slices.",
    "instructions": "Toast the bread. Spread peanut butter. Add banana slices.",
    "servings": 1,
    "prep_time_minutes": 5,
    "cook_time_minutes": 3,
    "difficulty_level": "easy",
    "category_id": 1,
    "id": 3
  },
  {
```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Response headers

```
content-length: 1094
content-type: application/json
date: Fri,13 Mar 2026 20:13:23 GMT
server: uvicorn
```

Example Outputs / Demo

vegan_only
boolean | (boolean | null)
(query)

true

limit
integer
(query)

5

maximum: 20
minimum: 1

Filter for top 5 highest protein recipes

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/analytics/recipes/top-protein?limit=5' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/analytics/recipes/top-protein?limit=5

Server response

Code	Details
200	Response body


```
[
  {
    "recipe_id": 1,
    "recipe_name": "Chicken Rice Bowl",
    "value": 66.05
  },
  {
    "recipe_id": 11,
    "recipe_name": "Chicken Pasta Plate",
    "value": 64.61
  },
  {
    "recipe_id": 4,
    "recipe_name": "Salmon Potato Plate",
    "value": 52.36
  },
  {
    "recipe_id": 10,
    "recipe_name": "Cheese Omelette Toast",
    "value": 31.9
  },
  {
    "recipe_id": 2,
    "recipe_name": "Protein Oat Bowl",
```


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Response headers

```
content-length: 329
content-type: application/json
date: Fri,13 Mar 2026 20:15:03 GMT
server: uvicorn
```

Example Outputs / Demo

vegan_only
boolean | (boolean | null)
(query)

true

limit
integer
(query)

5

maximum: 20
minimum: 1

Filter for top 5 lowest cost recipes

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/analytics/recipes/lowest-cost?limit=5' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/analytics/recipes/lowest-cost?limit=5

Server response

Code	Details
200	

Response body


```
[
  {
    "recipe_id": 3,
    "recipe_name": "Peanut Banana Toast",
    "value": 0.53
  },
  {
    "recipe_id": 8,
    "recipe_name": "Chocolate Strawberry Bites",
    "value": 0.97
  },
  {
    "recipe_id": 12,
    "recipe_name": "Overnight Oats Cup",
    "value": 1.13
  },
  {
    "recipe_id": 10,
    "recipe_name": "Cheese Omelette Toast",
    "value": 1.18
  },
  {
    "recipe_id": 9,
    "recipe_name": "Tofu Salad Bowl",
```


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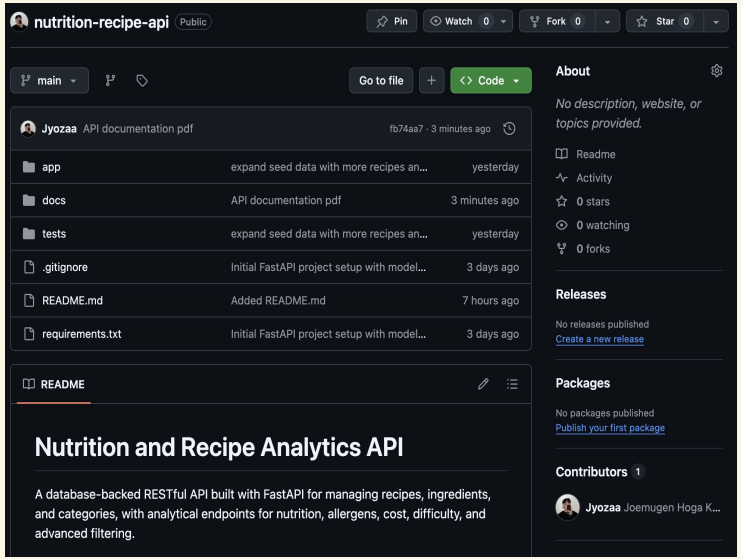
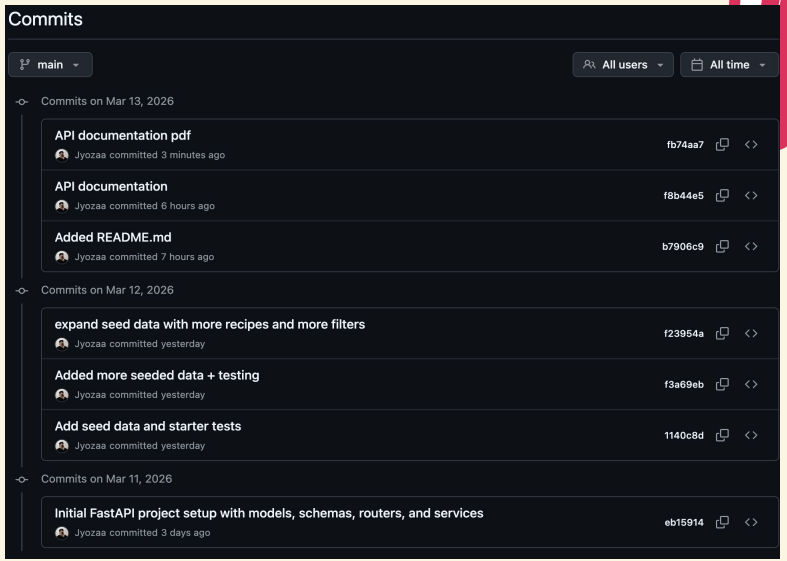
Response headers

```
content-length: 332
content-type: application/json
date: Fri,13 Mar 2026 20:16:47 GMT
server: uvicorn
```

Testing + Version Control



Github repo



Commit history

Testing + Version Control

```
(.venv) Joemugens-MacBook-Pro:nutrition-recipe-api joe$ PYTHONPATH=. pytest tests/test_recipes.py -v
========================================================== test session starts ==========================================================
platform darwin -- Python 3.13.12, pytest-8.4.1, pluggy-1.6.0 -- /Users/joe/Desktop/nutrition-recipe-api/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/joe/Desktop/nutrition-recipe-api
plugins: anyio-4.12.1
collected 8 items

tests/test_recipes.py::test_create_recipe PASSED                                                                                [ 12%]
tests/test_recipes.py::test_create_recipe_with_invalid_category PASSED                                                          [ 25%]
tests/test_recipes.py::test_add_duplicate_recipe_ingredient_rejected PASSED                                                     [ 37%]
tests/test_recipes.py::test_recipe_detail_contains_ingredients PASSED                                                           [ 50%]
tests/test_recipes.py::test_filter_recipes_by_vegan_only PASSED                                                                 [ 62%]
tests/test_recipes.py::test_filter_recipes_by_gluten_free_only PASSED                                                           [ 75%]
tests/test_recipes.py::test_filter_recipes_by_max_calories PASSED                                                               [ 87%]
tests/test_recipes.py::test_filter_recipes_by_max_cost PASSED                                                                   [100%]

========================================================== 8 passed in 0.12s ==========================================================
```

```
(.venv) Joemugens-MacBook-Pro:nutrition-recipe-api joe$ PYTHONPATH=. pytest
========================================================== test session starts ==========================================================
platform darwin -- Python 3.13.12, pytest-8.4.1, pluggy-1.6.0
rootdir: /Users/joe/Desktop/nutrition-recipe-api
plugins: anyio-4.12.1
collected 17 items

tests/test_analytics.py ....                                                                                                    [ 23%]
tests/test_categories.py ...                                                                                                    [ 41%]
tests/test_ingredients.py ..                                                                                                    [ 52%]
tests/test_recipes.py ........                                                                                                  [100%]

========================================================== 17 passed in 0.19s ==========================================================
```

Pytest passing

Reflection and Future Work

Challenges

many-to-many schema design
dynamic analytics logic
test database setup
balancing scope and polish

Future work

public nutrition dataset
integration
authentication
deployment
pagination and richer analytics

Thanks!